

INFATON MCP35: 35 Tools to Connect AI Agents with 1C:Enterprise ERP

TL;DR: We open-sourced an MCP server that gives AI assistants full access to 1C:Enterprise ERP — the most widely used ERP in Russia (300,000+ organizations). 35 tools, pure BSL, JSON-RPC 2.0. [Git-Hub repo](https://github.com/infaton/MCP35) (<https://github.com/infaton/MCP35>).

The Problem

1C:Enterprise is the backbone of Russian business automation — accounting, warehousing, manufacturing, HR, payroll. But it’s a closed ecosystem with its own language (BSL), its own IDE, and virtually zero AI integration.

Until now, connecting an AI agent to a 1C database meant building custom REST APIs for every single operation. Want to read a document? Write an API. Query a register? Another API. Check metadata? Yet another one.

The Solution: Model Context Protocol

We built **INFATON MCP35** — a native BSL HTTP service that implements the [Model Context Protocol](https://modelcontextprotocol.io/) (<https://modelcontextprotocol.io/>) specification (JSON-RPC 2.0). One deployment gives your AI assistant **35 tools** to work with any 1C database.

What Can It Do?

Metadata & Structure (6 tools)

Tool	Description
<code>get_metadata_tree</code>	Full configuration metadata tree
<code>get_metadata_details</code>	Detailed object metadata
<code>get_object_module</code>	BSL source code of any module
<code>list_subsystems</code>	Subsystem hierarchy
<code>list_common_modules</code>	Common modules catalog
<code>list_enums</code>	Enumerations and values

Reference Data (6 tools)

Tool	Description
<code>list_catalogs</code>	All catalogs in config
<code>get_catalog_items</code>	Read catalog items with filters
<code>get_catalog_item</code>	Single item by ID
<code>create_catalog_item</code>	Create new item
<code>update_catalog_item</code>	Update existing
<code>search_catalog</code>	Full-text search

Documents (7 tools)

Tool	Description
<code>list_documents</code>	All document types
<code>get_documents</code>	Query with filters/date range
<code>get_document</code>	Single document
<code>create_document</code>	Create new
<code>update_document</code>	Modify existing
<code>post_document</code>	Post (conduct)
<code>unpost_document</code>	Unpost

Registers (5 tools)

Accumulation, information, and accounting registers — read balances, turnovers, movements, slices.

BSP Integration (6 tools)

Built-in Subsystem Library: users, access rights, print forms, attached files, additional properties.

Queries & Utils (5 tools)

`execute_query` (raw 1C query language), scheduled jobs, event log, data locks, version info.

Architecture

```
AI Client (Claude/GPT/local LLM)
  ↓ JSON-RPC 2.0
INFATON MCP Server (BSL HTTP Service)
  ↓ COM/Native API
1C:Enterprise Database
```

The server runs as an HTTP service inside 1C:Enterprise 8.3.25+. No external dependencies, no Node.js, no Python — pure BSL.

Quick Start

1. Download `INFATON_MCP.cfe` from [Releases](https://github.com/infaton/MCP35/releases) (<https://github.com/infaton/MCP35/releases>)
2. Load the extension in 1C Configurator
3. Publish the HTTP service
4. Point your MCP client to `http://your-1c-server/your-base/hs/mcp`

Why Open Source?

MCP35 is the **1C-side** component of the [INFATON platform](https://infaton.ru) (<https://infaton.ru>) — an AI-powered ERP implementation toolkit featuring:

- **Automatic survey cycle** with 3 discovery engines
- **Digital twins** of ERP consultants in 3 modes: Advisor, Autopilot, or Both
- **Airgapped mode** for classified environments with local LLMs only
- **Problem tracing** (INFATON SA-8 methodology)

We open-sourced the MCP server because we believe every 1C database should be AI-accessible. The platform adds the intelligence layer on top.

Links

- **GitHub:** github.com/infaton/MCP35 (<https://github.com/infaton/MCP35>)
- **Website:** infaton.ru (<https://infaton.ru>)
- **Build instructions:** [HOW_TO_BUILD_CFE.md](https://github.com/infaton/MCP35/blob/master/HOW_TO_BUILD_CFE.md) ([https://github.com/infaton/MCP35/blob/master/](https://github.com/infaton/MCP35/blob/master/HOW_TO_BUILD_CFE.md)
`HOW_TO_BUILD_CFE.md`)

Tags: #mcp #1c #erp #ai #opensource #bsl #json-rpc #digital-twin